

Ashley Nielsen

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EDUCATION

Brigham Young University

Bachelor of Science: Machine Learning

Minor in Mathematics

Expected December 2027

Provo, UT

- GPA: 3.97
- Relevant Coursework: CS 473: Advanced Machine Learning, CS 312: Algorithm Design and Analysis, CS 240: Software Design and Construction

SKILLS

Languages: Python, Kotlin, Java, C++, SQL

ML / Data: PyTorch, NumPy, pandas, scikit-learn, matplotlib, seaborn

Tools: AWS (DynamoDB, API Gateway, Cloudwatch), Git

EXPERIENCE

Church of Jesus Christ of Latter-day Saints

April 2025-January 2026

Technical Intern

Riverton, UT

- Built an internal error-reporting tool that reduced debugging time by an average of 20% across 5+ team members
- Created an AWS DynamoDB table with high-speed access that stored 1M+ records
- Collaborated with QA engineers to validate fixes and improve overall system reliability

CLASS PROJECTS

Chess Game

- Built a full-stack networked chess application in Java with a command-line client, REST API server, and shared modules for game state management
- Implemented validation logic ensuring 100% of games were played using legal moves

Machine Learning Model Comparison

- Trained and evaluated multiple ML models on structured datasets with 99% test accuracy
- Compared models using accuracy-based metrics and analyzed overfitting and generalization behavior

The Number Game

- Implemented a Bayesian-inference model that computed posterior probabilities for mathematical concepts (evens, squares, powers, etc.) given observed number sets
- Displayed graphs comparing prior and posterior probabilities using matplotlib

VOLUNTEER EXPERIENCE

Church of Jesus Christ of Latter-day Saints

June 2023-December 2024

Full-time Representative

Puebla, Mexico

- Trained 3 new volunteers, helping them adapt to the rigorous schedule, become accustomed to the culture, and improve their teaching abilities
- Led a group of 6-8 volunteers, followed up on goals, and maintained positive relationships